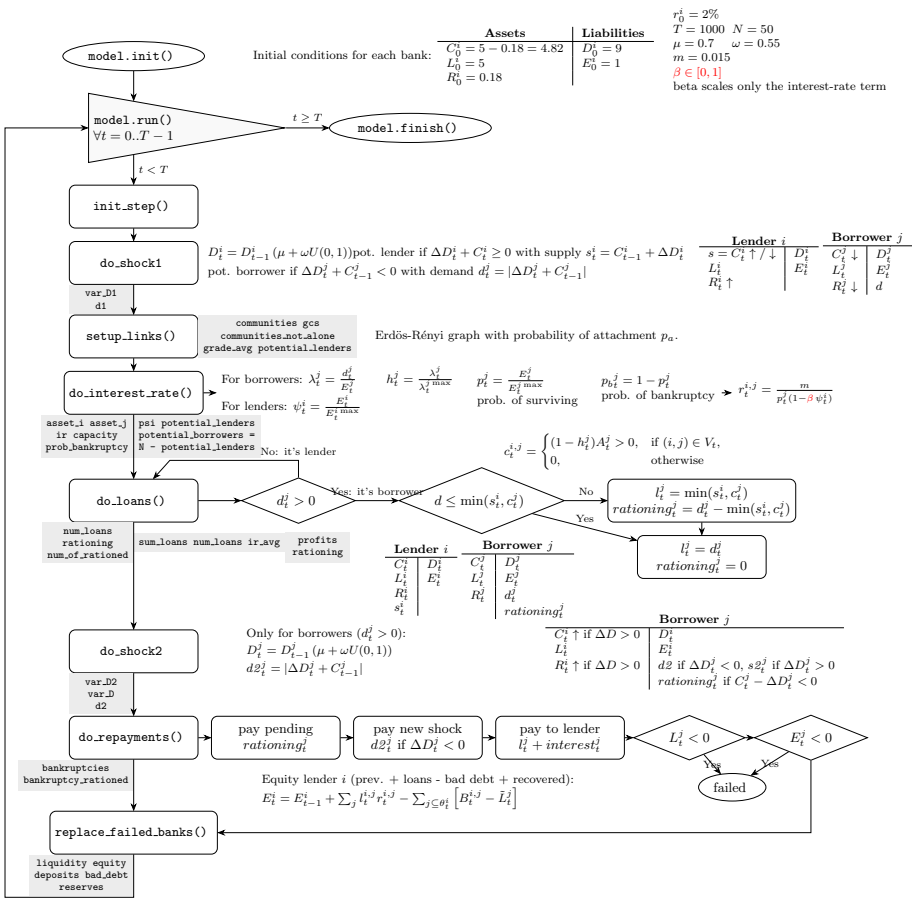

model.init()
Initial conditions for each bank:
Assets
Liabilities
$C_0^i = 5 - 0.18 = 4.82$
$L_0^i = 5$
$R_0^i = 0.18$
$D_0^i = 9$
$E_0^i = 1$
$r_0^i = 2\%$
$T = 1000 \quad N = 50$
$\mu = 0.7 \quad \omega = 0.55$
$m = 0.015$
$\beta \in [0,1]$
beta scales only the interest-rate term
model.run()
$\forall t = 0..T-1$
$t \geq T$
model.finish()
$t < T$
init_step()
do_shock1
$D_t^i = D_{t-1}^i\,(\mu + \omega U(0,1))$pot. lender if $\Delta D_t^i + C_t^i \geq 0$ with supply $s_t^i = C_{t-1}^i + \Delta D_t^i$
pot. borrower if $\Delta D_t^j + C_{t-1}^j < 0$ with demand $d_t^j = |\Delta D_t^j + C_{t-1}^j|$
Lender i
$s = C_t^i \uparrow / \downarrow$ | D_t^i
L_t^i | E_t^i
$R_t^i \uparrow$
Borrower j
$C_t^j \downarrow$ | D_t^j
L_t^j | E_t^j
$R_t^j \downarrow$ | d
var_D1
d1
setup_links()
communities gcs
communities_not_alone
grade_avg potential_lenders
Erdös-Rényi graph with probability of attachment p_a.
do_interest_rate()
For borrowers: $\lambda_t^j = \frac{d_t^j}{E_t^j}$
For lenders: $\psi_t^i = \frac{E_t^i}{E_t^{i\,\max}}$
$h_t^j = \frac{\lambda_t^j}{\lambda_t^{j\,\max}}$
$p_t^j = \frac{E_t^j}{E_t^{j\,\max}}$
prob. of surviving
$p_{b_t}^j = 1 - p_t^j$
prob. of bankruptcy
$r_t^{i,j} = \frac{m}{p_t^j(1-\beta\,\psi_t^i)}$
asset_i asset_j
ir capacity
prob_bankruptcy
psi potential_lenders
potential_borrowers =
N - potential_lenders
No: it's lender
$c_t^{i,j} = \begin{cases} (1-h_t^j)A_t^j > 0, & \text{if } (i,j) \in V_t, \\ 0, & \text{otherwise} \end{cases}$
do_loans()
$d_t^j > 0$
Yes: it's borrower
$d \leq \min(s_t^i, c_t^j)$
No
Yes
$l_t^j = \min(s_t^i, c_t^j)$
$rationing_t^j = d_t^j - \min(s_t^i, c_t^j)$
$l_t^j = d_t^j$
$rationing_t^j = 0$
num_loans
rationing
num_of_rationed
sum_loans num_loans ir_avg
profits
rationing
Lender i
C_t^i | D_t^i
L_t^i | E_t^i
R_t^i
s_t^i
Borrower j
C_t^j | D_t^j
L_t^j | E_t^j
R_t^j | d_t^j
$rationing_t^j$
do_shock2
Only for borrowers ($d_t^j > 0$):
$D_t^j = D_{t-1}^j\,(\mu + \omega U(0,1))$
$d2_t^j = |\Delta D_t^j + C_{t-1}^j|$
Borrower j
$C_t^j \uparrow$ if $\Delta D > 0$ | D_t^j
L_t^j | E_t^j
$R_t^j \uparrow$ if $\Delta D > 0$ | $d2$ if $\Delta D_t^j < 0$, $s2_t^j$ if $\Delta D_t^j > 0$
$rationing_t^j$ if $C_t^j - \Delta D_t^j < 0$
var_D2
var_D
d2
do_repayments()
pay pending $rationing_t^j$
pay new shock $d2_t^j$ if $\Delta D_t^j < 0$
pay to lender $l_t^j + interest_t^j$
$L_t^j < 0$
$E_t^j < 0$
Yes
Yes
failed
bankruptcies
bankruptcy_rationed
Equity lender i (prev. + loans - bad debt + recovered):
$E_t^i = E_{t-1}^i + \sum_j l_t^{i,j} r_t^{i,j} - \sum_{j \subseteq \theta_t^i} \left[B_t^{i,j} - \hat{L}_t^j \right]$
replace_failed_banks()
liquidity equity
deposits bad_debt
reserves